

# Rory McCulloch

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Seeking Summer 2027 international co-op opportunities in AI/ML and embedded systems.

## SUMMARY

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Dual-degree Electrical Engineering & Computing Technology student at uOttawa, currently interning at Trench Group as an AI/ML Applied Sciences Intern on Project NOVA. Developing ML-driven pipelines to modernize data aggregation systems for high-voltage equipment including instrument transformers and bushings.

## PROFESSIONAL EXPERIENCE

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### AI/ML Applied Sciences Intern, Trench Group – Project NOVA

May '26 – Oct. '26

- Developing ML models to modernize legacy data aggregation and collection pipelines for high-voltage equipment including instrument transformers and bushings.
- Building and deploying intelligent, scalable applications in collaboration with cross-functional engineering teams and senior leadership to drive operational insight.

### Electrical Engineering Intern, Aim Colours

May '25 – Sep. '25

- Designed and routed multi-layer PCB backplanes for wearable technology, collaborating with industrial designers to ensure components fit strict ergonomic form factors.
- Analyzed power consumption profiles and collated simulation data for embedded microcontrollers (XIAO MG24), applying strong debugging skills to optimize circuit design.
- Streamlined DFM validation workflows and performed rigorous hardware verification, significantly reducing production errors and improving overall prototype quality.

## PROJECTS & LEADERSHIP

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### Counter-UAV System, Shazam4Drones – Defense Tech

Nov. '25

- Built real-time drone detection by audio, achieving 70% accuracy in a national defence hackathon placing 4th out of 87 teams.
- Engineered a deployable counter-UAV pipeline integrating audio classification with a real-time signal processing backend for military field use.
- Prototyped a battlefield mobile app interface for the detection system, later adapted into a standalone embedded device for improved field practicality.

### Co-Lead, Student Professional Awareness Conference (SPAC)

Oct. '25

- Organized a conference of 300+ attendees for networking between students and professionals.
- Raised thousands of dollars in sponsorship by securing 20+ industry partners.
- Coordinated several hands-on workshops in parallel with networking and panelist presentations.

### Live Weather Tracking System for Aircraft, uOttHack

Jan. '25

- Used an Arduino & temperature sensor to develop a live weather tracking system for aircrafts.
- Created an appealing front-end using Chart.js to display a live graph accompanied by user-friendly UI made with CSS and HTML.
- Bridged the back-end running on Python to the front-end using NPM to host a server.

### VP External, IEEE uOttawa Student Branch

Sep. '24 – Present

- Secured 30+ industry sponsors and partners, leading outreach strategy and managing two Industry Coordinators to drive corporate engagement for branch events.
- Organized a Ciena Networking Night connecting 200+ students with 20 Ciena engineers and interns for structured professional networking.
- Hosted a Resume Roast panel bringing in interns from Microsoft, Tesla, Meta, AMD, Nokia, and P&G to deliver live resume feedback to students.

## UNIVERSITY OF OTTAWA

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### BASc in Electrical Engineering and BSc in Computing Technology

Sep. '24 – Apr. '29

Computer Architecture, Circuit Theory I & II, Data Structures and Algorithms

## SKILLS

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### Languages

Python, C++, Java, Go, R, JavaScript

### AI/ML & Data

PyTorch, scikit-learn, NumPy, pandas

### Hardware & Tools

KiCad, Arduino, Visual Studio Code, Git

### Spoken Languages

English (native), French (bilingual)